

Beginner's Guide: NGC 4429

ALMA Thesis Planner (automated pipeline)

0.1 1. The Goal

At the very center of almost every big galaxy sits a supermassive black hole — an object so dense that not even light can escape it once it gets too close, with a mass that can be millions or billions of times that of our Sun. Around some of these black holes, astronomers have noticed something strange: a "hole" or empty-looking gap in the gas that swirls around them, like a doughnut hole in the middle of what should otherwise be a solid disk of material. This gap shows up when astronomers look at carbon monoxide gas (called "CO" — it's used because it's the easiest way to trace hydrogen gas, which is hard to see directly). Nobody is sure whether this gap is a *true* empty cavity — meaning the black hole (or some explosive activity around it) has actually blown or eaten away all the gas — or whether it's a *fake* gap, meaning there's still gas there, just not the kind of gas that CO happens to be good at lighting up.

This project looks at one specific galaxy, NGC 4429, using ALMA (a giant array of radio telescopes in Chile) data that has already been collected and calibrated — you don't need to observe anything new. You have two "pictures" of the galaxy's center: one is a 3D data cube (like a stack of images, one for each narrow slice of velocity) of a molecule called HCN, which only lights up in much denser gas than CO does. If HCN also shows a gap, that's stronger evidence the hole is genuinely empty of gas. If HCN doesn't show a gap (or you can't tell either way), it suggests CO was just the wrong tracer to see gas that's actually still there. The second picture is a continuum image (a simple "photo" at radio wavelengths, without splitting light into different velocities) that can reveal whether there's a compact bright dot right at the black hole's location — a hint of activity there — or nothing unusual.

Why should you care? This is a real, unanswered piece of a bigger mystery: what shapes the immediate surroundings of a supermassive black hole, and does the black hole's own energy output ("feedback") clear out its neighborhood, or is it just gravity quietly doing its thing over billions of years? You would be one of the first people to actually look at this specific dataset (it hasn't been published anywhere yet, per Semantic Scholar).

By the end, you should have: (1) a set of maps of the HCN gas in the galaxy's center showing how much of it there is, how fast it's moving, and how spread out in velocity it is; (2) a plot of gas brightness versus distance from the black hole, showing whether there's a dip in the middle or not; (3) either a real measurement of how much dense gas is there, or — more likely — a solid upper limit ("there's less than this much gas") with a clearly stated range depending on assumptions; (4) a description of whether there's a compact radio source right at the black hole; and (5) a short written verdict, in your own words, on whether the evidence favors "real empty hole" or "hole that's just invisible to CO" — with honest caveats about what your data can't tell you.

0.2 2. Skills You'll Need

Python, at a basic-to-intermediate level. You'll use it to open astronomy image files, do simple math and plotting, and run a few specialized astronomy packages. You don't need to be a "programmer" — you need to be comfortable writing short scripts, using loops, and calling functions from libraries other people wrote. If you're new to Python, a beginner course (e.g., a free intro-to-Python course, or the first few chapters of any "Python for data analysis" book) covering variables, loops, functions, and the NumPy/Matplotlib libraries is enough to start. You'll pick up the astronomy-specific packages (astropy, spectral-cube, radio-beam, photutils) as you go — they have their own tutorials and documentation with worked examples.

Basic astrophysics concepts. You should understand, in simple terms:

- What a galaxy is, and specifically what a "molecular gas disk" is — a flat, rotating disk of cold gas and dust in a galaxy, similar to how planets orbit in a flat disk around a star.
- What a supermassive black hole is and why galaxies have one at the center.
- What "radio telescope data" and a "spectral line cube" are — think of the cube like a loaf of bread sliced very thin, where each slice is a picture of the galaxy at one specific velocity (astronomers use the Doppler effect — the same idea as an ambulance siren changing pitch — to map how gas is moving).
- What "flux" means (basically brightness — how much light/radio energy you're receiving) and what "distance" means for converting an angular size on the sky (arcseconds) into a physical size (light-years or parsecs) in the galaxy itself.
- What "resolution" or "beam size" means — essentially, the blurriness limit of the telescope; anything smaller than the beam gets smeared out and can't be seen as separate detail. This is the single most important concept for this project, since the whole "is it a real hole" question hinges on whether the suspected gap is bigger or smaller than what the telescope can resolve.

None of this requires calculus — arithmetic, basic algebra, and comfort with logarithms (since astronomy numbers span huge ranges) is enough. If these are new to you, a free "intro to astronomy" course or the first few chapters of any general astronomy textbook (the kind used in a non-major "intro to the universe" college course) will cover galaxies, black holes, and basic radio astronomy ideas.

Statistics, lightly. You'll need to understand what "noise" and "signal-to-noise ratio" mean (is this bump in the data real, or could it just be random static?), and how to compute a simple upper limit when you don't detect something. This is usually taught in the first weeks of any intro-to-astronomy-research or intro stats course, and you'll also just learn it by doing this project.

Tools: No CASA installation is strictly required — everything can be done in Python with 'astropy', 'spectral-cube', 'radio-beam', and 'photutils', which are the beginner-friendlier route. CASA (the specialized radio-astronomy software) is an alternative for the same steps, but is heavier to install and has a steeper learning curve, so start with Python.

0.3 3. Your First Week

Day 1 — Get oriented, don't touch data yet. Read background on supermassive black holes, molecular gas disks, and what a "circumnuclear hole" is (search a couple of review-style astronomy articles or an intro textbook chapter). Read through the actual proposal abstract and

the data description you were given, twice — the goal today is just to be able to explain, in your own words, out loud, what question this project is trying to answer and why the HCN vs. CO distinction matters. Install Python (if you haven't) via Anaconda, and install 'astropy', 'spectral-cube', 'radio-beam', 'photutils', 'numpy', and 'matplotlib'.

Day 2 — Open the data and just look at it. Load the HCN cube and the continuum image with 'astropy.io.fits'. Print the header information — this is your first real "research" moment, matching up numbers in the header (pixel scale, beam size, rest frequency, number of channels) against what's described in your data summary. Use 'matplotlib' to just display a single channel of the cube as an image, and separately display the continuum image. Don't worry about doing anything scientific yet — just confirm you can see the galaxy and that the images look reasonable (a bright blob near the center, roughly what you'd expect). Expect some frustration with WCS (World Coordinate System) axes and FITS conventions — this is normal on day one of any radio-astronomy project.

Day 3 — Explore the cube in velocity. Extract a spectrum (brightness vs. velocity/channel) at the pixel closest to the black hole's coordinates, and at a few random off-center pixels. Plot these. This is how you'll visually identify which channels contain real HCN emission (a bump) versus which are just noise (flat, jittery). This is also a good day to read a short tutorial on Doppler shifting and radial velocity, since it will make the x-axis of your spectrum plots make sense.

Day 4 — Measure the noise. Using the line-free channels you identified on Day 3, compute the RMS (root-mean-square) noise per channel across the cube, restricted to the central region within the primary beam. Plot this as a curve across all 126 channels. This is the "feasibility check" — by the end of today you should be able to say, roughly, "yes, if there's HCN emission of at least X brightness, we could see it" or "no, this data is too noisy to say much." Expect this step to take longer than it sounds; getting masking/region selection right is fiddly the first time.

Day 5 — First moment map. Using your noise curve from Day 4 to set a detection threshold (e.g., only include channels/pixels above $3\times$ the noise), make your first moment-0 map (a 2D image of total HCN brightness, collapsing all velocity channels into one picture) using 'spectral_cube's built-in 'moment()' function. Don't aim for the polished nuclear zoom-in yet — just get any moment map to appear, sanity-check it against the raw channel images from Day 2, and celebrate that milestone.

End of week 1, realistically: You should have a working Python environment, both FITS files loaded and inspected, a noise curve for the HCN cube, and a first-draft (rough, not final) moment-0 map. You will likely *not* yet have moment-1/2 maps, the radial profile, or any upper-limit numbers — those are reasonable goals for week 2. Getting stuck on WCS/coordinate conversions or on picking the right line-free channels is completely normal at this stage; if you're stuck for more than half a day on an installation or plotting quirk, that's the moment to ask your advisor or search the package's GitHub issues rather than pushing through alone.